

## Scalability & Future Plan

**Date:** 15 March 2026

**Team ID:** (xxxx)

**Project Name:** Human Development Index (HDI) Prediction Using Machine Learning

**Maximum Marks:** 1 Mark

---

### Scalability & Future Plan

The **Human Development Index (HDI) Prediction Using Machine Learning** project is designed with a scalable architecture using **Python, Flask, Scikit-learn, Pandas, NumPy, HTML, CSS, and JavaScript**. The current system provides accurate HDI predictions based on key development indicators. In the future, the application can be enhanced to support larger datasets, more users, cloud deployment, and additional analytical features to improve performance and usability.

---

### Current System Limitations:-

| S.No | Limitation                                                 | Impact                                                   | Priority to Address |
|------|------------------------------------------------------------|----------------------------------------------------------|---------------------|
| 1    | Application is hosted on a local Flask development server. | Cannot support many concurrent users or internet access. | High                |
| 2    | Uses a static CSV dataset for prediction.                  | Data may become outdated over time.                      | High                |
| 3    | No user authentication or role-based access control.       | Limited security for future enterprise use.              | Medium              |

---

### Scalability Plan:-

| S.No | Scalability Aspect | Current State                | Proposed Upgrade / Solution                                                                                   |
|------|--------------------|------------------------------|---------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports local users only.   | Deploy on cloud platforms such as Render, Railway, or AWS to support multiple users simultaneously.           |
| 2    | Data Storage       | Stores data in CSV files.    | Migrate to a relational database such as MySQL or PostgreSQL for better scalability and data management.      |
| 3    | Performance        | Single machine prediction.   | Optimize the machine learning model, implement caching, and use a production WSGI server (Gunicorn/Waitress). |
| 4    | Security           | Basic input validation only. | Add user authentication, HTTPS, encrypted data transfer, and role-based access control.                       |

#### Future Roadmap:-

| Phase          | Planned Feature / Enhancement                                                | Target Timeline | Expected Impact                                                  |
|----------------|------------------------------------------------------------------------------|-----------------|------------------------------------------------------------------|
| <b>Phase 1</b> | Deploy the Flask application to a cloud hosting platform.                    | 1–2 Months      | Makes the application accessible online for users from anywhere. |
| <b>Phase 2</b> | Integrate real-time HDI datasets from trusted public sources.                | 3–4 Months      | Improves prediction accuracy using updated information.          |
| <b>Phase 3</b> | Add interactive dashboards, graphs, and country comparison features.         | 4–6 Months      | Enhances data visualization and decision-making capabilities.    |
| <b>Phase 4</b> | Implement user authentication, report generation, and mobile responsiveness. | 6–8 Months      | Improves security, usability, and overall user experience.       |